Homework 1 Statistical Machine Learning

Ville Wassberg

November 2024

Problem 1

In this problem we are given two one-armed bandits B_1 and B_2. B_1 is run n_1 times with rewards $R_1^1, \ldots, R_1^{n_1}$ and B_2 is run n_2 times with rewards $R_2^1, \ldots, R_2^{n_2}$. We let the data be represented by $X = \{R_1^1, \ldots, R_1^{n_1},\ R_2^1, \ldots, R_2^{n_2}\}$. We have the action space $\aleph = \{0,\ 1\}$ and let $Y = (R_1, R_2)$, and loss functions $L_1(Y, a) = -R_a$ and $L_2(Y, a) = 1_{R_a=0}$.

To find the formal Bayes rules $\delta_i(x)$ we aim to minimise the Bayesian posterior risk function: $\min_{a\in\aleph} \mathbb{E}[L_i(Y, \delta(x) = a)|X = x]$, $i = 1, 2$. This is quite straight forward and we get the following:

$$\delta_1(x) = \arg\min_{a\in\aleph} \mathbb{E}[-R_a|X = x] = \begin{cases} 1, & \text{if } \mathbb{E}[R_1|X = x] \geq \mathbb{E}[R_2|X = x], \\ 2, & \text{else,} \end{cases}$$

$$\delta_2(x) = \arg\min_{a\in\aleph} \mathbb{E}[R_a = 0|X = x] = \begin{cases} 1, & \text{if } 1 \cdot \Pr(R_1 = 0|X = x) \leq 1 \cdot \Pr(R_2 = 0|X = x), \\ 2, & \text{else.} \end{cases}$$

Problem 2

We are given the following distributions:

$$Z_a^i|\theta_i \sim \mathrm{Ber}(\theta_a),\ Z_a^i = 1_{\{R_a^i=0\}},\ R_a^j|(Z_a^j = 1, \lambda_a) \sim \mathrm{Exp}(\lambda_a),$$
$$\theta_a \sim \mathrm{Beta}(\alpha, \beta),\ \lambda_a \sim \Gamma(\gamma, \delta),\ \ \Pr(R_a \neq 0) = \theta_a.$$

as well as that all data are iid given the parameters. To find the formal Bayes rule from problem 1 as a function of x, γ, β, α and δ oen can make use of the predictive posterior formula $p(Y|X) = \int p(Y|\theta)\, p(\theta|X)\, d\theta$, thus, the predictive posterior of this problem can be described as

$$p_{R_a|X}(r|x) = \int p_{R_a|\theta_a,\lambda_a,Z_a=1,X}(r|\theta_a, \lambda_a, Z_a = 1, x)\, p_{\theta_a,\lambda_a,Z_a=1|X}(\theta_a, \lambda_a, Z_a = 1|x)\, d\theta_a\, d\lambda_a.$$

We can begin by notice that

$$\mathbb{E}[R_a|X=x] = \mathbb{E}[R_a|X=x, Z_a=1]\Pr(Z_a=1) + \mathbb{E}[R_a|X=x, Z_a=0]\Pr(Z_a=0) = \mathbb{E}[R_a|X=x, Z_a=1]\theta_a,$$

assuming the rewards cannot be negative. Thus we only need to care about when $Z_a = 1$. Here Z_a and R_a are to be understood as $Z_a^{n_a+1}$ and $R_a^{n_a+1}$. This together with the data being iid given the parameters allows us to drop θ_a as well as X in $p_{R_a|\theta_a,\lambda_a,Z_a=1,X}(r|\theta_a, \lambda_a, Z_a = 1, x)$.

Due to iid of data and that λ_a only affect the distribution of R_a given $Z_a = 1$ we can write the second density in the predictive posterior as

$$p_{\theta_a,\lambda_a,Z_a=1|X}(\theta_a, \lambda_a, Z_a = 1|x) = p_{Z_a=1|\theta_a,\lambda_a,X}(Z_a = 1|\theta_a, \lambda_a, x)\, p_{\theta_a,\lambda|X}(\theta_a, \lambda|x) = p_{Z_a=1|\theta_a}(Z_a = 1|\theta_a)\, p_{\theta_a|X}(\theta_a|x)\, p_{\lambda_a|X}(\lambda_a|x).$$

Now, the predictive posterior can be described as

$$p_{R_a|X}(r|x) = \int p_{R_a|Z_a=1,\lambda_a}(r|Z_a = 1, \lambda_a)\, p_{Z_a=1|\theta_a}(Z_a = 1|\theta_a)\, p_{\theta_a|X}(\theta_a|x)\, p_{\lambda_a|X}(\lambda_a|x)\, d\theta_a\, d\lambda_a.$$

Hence, the unknowns to be found are the densities of $\theta_a|X$ and $\lambda_a|X$. Beginning with $\theta_a|X$:

Letting $Z_a^{1:n_a}$ be the vector of $Z_a^1, \dots, Z_a^{n_a}$ and recalling data is iid given the parameters we have the posterior

$$p_{\theta_a|Z_a^{1:n_a}}(\theta_a|Z_a^{1:n_a}) \propto p_{Z_a^{1:n_a}|\theta_a}(Z_a^{1:n_a}|\theta_a)p_{\theta_a}(\theta_a) = \Pi_i^{n_a} p_{Z_a^i|\theta_a}(Z_a^i|\theta_a)p_{\theta_a}(\theta_a) \propto \prod_{i=1}^{n_a} \theta_a^{Z_a^i}(1-\theta_a)^{1-Z_a^i}\theta_a^{\alpha-1}(1-\theta_a)^{\beta-1} = \theta_a^{\sum_{i=1}^{n_a} Z_a^i+\alpha-1}(1-\theta_a)^{n_a-\sum_{i=1}^{n_a} Z_a^i+\beta-1},$$

which can be recognised being proportional to a beta distribution. Letting $S_a = \sum_{i=1}^{n_a}$ and since Z_a is merely an indicator of the data X, we have

$$\theta_a|X \sim \mathrm{Beta}_{\theta_a}\left(S_a + \alpha, n_a - S_a + \beta\right).$$

For $\lambda_a|X$: We do the same here. For the likelihood, $X|\lambda_a = X|Z_a^{1:n} = 1, \lambda_a$ we have to consider that we only care for when the reward is positive, hence, $p_{X|Z_a^{1:n}=1,\lambda_a} = \prod_{i=1}^{S_a} \lambda_a e^{-\lambda_a R_a^i} = \lambda_a^{S_a} e^{-\lambda_a \sum_{i=1}^{S_a} R_a^i}$. Thus, $\lambda_a|X$ is distributed by

$$p_{\lambda|X}(\lambda|x) \propto p_{x|\lambda}(x|\lambda)p_\lambda(\lambda) = \lambda_a^{\gamma-1+S_a} e^{-\lambda_a(\delta+\sum_{i=1}^{S_a} R_a^i)},$$

which can be recognised as proportional to a gamma distribution

$$\lambda_a|X \sim \Gamma_{\lambda_a}\left(\gamma + S_a, \delta + \sum_{i=1}^{S_a} R_a^i\right).$$

We now have enough to compute the predictive posterior

$$
\begin{aligned}
p_{R_a|X}(r|x) = \\
\int \lambda_a e^{-\lambda_a R_a}\, \theta_a^{Z_a}(1-\theta_a)^{1-Z_a}\, \theta_a^{S_a+\alpha-1}(1-\theta_a)^{n_a-S_a+\beta-1}\, \lambda_a^{\gamma-1+S_a} e^{-\lambda_a(\delta+\sum_{i=1}^{S_a} R_a^i)}\, d\theta_a\, d\lambda_a = \\
\int_0^\infty \lambda_a^{\gamma+S_a} e^{-\lambda_a(R_a+\delta+\sum_{i=1}^{S_a} R_a^i)}\, d\lambda_a \int_0^1 \theta_a^{Z_a+S_a+\alpha-1}(1-\theta_a)^{n_a+1-Z_a-S_a+\beta-1}\, d\theta_a = \\
\frac{\Gamma(\gamma+S_a+1)}{(R_a+\delta+\sum_{i=1}^{S_a} R_a^i)^{\gamma+S_a+1}}\, B(Z_a+S_a+\alpha, n_a+1-Z_a-S_a+\beta),
\end{aligned}
$$

using that $B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}dt$ and $\Gamma(k) = \int_0^\infty t^{k-1}e^{-t}dt$. Thus, we get the formal Bayes rule $\delta_1(x)$ as

$$
\begin{aligned}
\mathbb{E}[R_a|X=x, Z_a=1]\theta_a = \int R_a\, p_{R_a|X}(r|x)\theta_a\, dR_a \\
\theta_a\, \Gamma(\gamma+S_a+1)\, B(Z_a+S_a+\alpha, n_a+1-Z_a-S_a+\beta) \int_0^\infty \frac{R_a}{(R_a+\delta+\sum_{i=1}^{S_a} R_a^i)^{\gamma+S_a+1}}\, dR_a.
\end{aligned}
$$

Letting $C = \delta+\sum_{i=1}^{S_a} R_a^i$ and $k = \gamma+S_a+1$, the integration part becomes

$$
\begin{aligned}
\int_0^\infty \frac{R_a}{(R_a+C)^k}\, dR_a = \int_C^\infty \frac{u-C}{u^k}\, du = \left[\frac{u^{2-k}}{(2-k)} + C\frac{u^{1-k}}{k-1}\right]_C^\infty = \\
\begin{cases} -C^{2-k}, & if S_a+\gamma > 1, \\ \infty, & else. \end{cases}
\end{aligned}
$$

Thus,

$$
\delta_1(x) = \arg\min_{a\in\aleph} \theta_a\, \Gamma(\gamma+S_a+a)\, B(Z_a+S_a+\alpha, n_a+1-Z_a-S_a+\beta)(-(\delta+\sum_{i=1}^{S_a} R_a^i))^{1-(\gamma+S_a)}.
$$

For $\delta_2(x)$ it is quite straight forward since we got

$$
\delta_2(x) = \arg\min_{a\in\aleph} \mathbb{E}[R_a = 0|X=x] = \begin{cases} 1, & \text{if } \Pr(R_1 = 0|X=x) \le \Pr(R_2=0|X=x), \\ 2, & \text{else.} \end{cases}
$$

Since,

$$
\begin{aligned}
\Pr(R_a = 0|X=x) = 1 - \Pr(Z_a = 1|X=x) = 1 - \int p_{Z_a=1|\Theta}(Z_a=1|x)\, p_\Theta(\theta)\, d\theta = \\
1 - \int \theta_a\, p_\Theta(\theta)\, d\theta = 1 - \mathbb{E}[\theta_a|X=x],
\end{aligned}
$$

and that we from previous computations got $\theta_a|X \sim \text{Beta}_{\theta_a}(S_a+\alpha, n_a - S_a + \beta)$, we get

$$\Pr(R_a = 0|X = x) = 1 - \frac{S_a+\alpha}{S_a+\alpha+n_a-S_a+\beta} = 1 - \frac{S_a+\alpha}{\alpha+n_a+\beta}.$$

Thus,

$$\delta_2(x) = \arg\max_{a\in\aleph} \frac{S_a+\alpha}{\alpha+n_a+\beta}.$$

Problem 3

Her we are given data $X = \{X_i\}_{i=1}^n$, $X_i \sim \text{Ber}(\theta)$ conditionally independent given $\Theta = \theta$. A prior on Θ such that $p_\Theta(\theta) \propto \frac{e^\theta}{e-1} \cdot \mathbf{1}\{0 \le \theta \le 1\}$. And the statistics $n_1 = \sum_{i=1}^n X_i$ and $n_0 = n - n_1$.

To make a Laplace approximation $q_{\Theta|X}(\theta|X) \sim \mathcal{N}(\theta^{MAP}, A^{-1})$ of the posterior, p, where $A = \frac{\partial^2}{\partial\theta^2} l(\theta)|_{\theta=\theta^{MAP}}$ we begin by finding maximum a posteriori. We have

$$p_{\Theta|X}(\theta \mid x) \propto p_X(x \mid \theta)p_\Theta(\theta) \propto \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i}\frac{e^\theta}{e-1}\cdot \mathbf{1}\{0\le\theta\le1\} \propto$$
$$\theta^{n_1}(1-\theta)^{n_0}e^\theta \cdot \mathbf{1}\{0\le\theta\le1\},$$

hence, a negative log likelihood

$$l(\theta) = -\log p_{\Theta|X}(\theta \mid x) = -\log\left(\theta^{n_1}(1-\theta)^{n_0}e^\theta\right) + \mathit{constant} = -\log p_{\Theta|X}(\theta \mid x) =$$
$$-n_0\log(1-\theta) - n_1\log\theta - \theta + \mathit{constant}, \quad \text{for } \theta \in [0,1].$$

Since maximising the posterior is equivalent to minimising $l(\theta)$ we find the zeros of the differential

$$\frac{\partial}{\partial\theta} l(\theta) = -\frac{n_1}{\theta} + \frac{n_0}{1-\theta} - 1 = 0 \text{ and } \theta \in [0,1]$$
$$\text{and } (n_1, n_0) = (3,7) \Rightarrow \theta^{MAP} = \frac{\sqrt{93}-9}{2}.$$

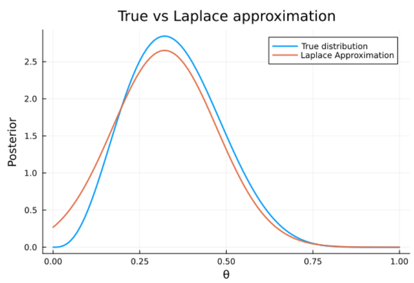
Now to find A we have

$$A = \frac{\partial^2}{\partial\theta^2} l(\theta)|_{\theta=\theta^{MAP}} = \frac{n_1}{(\theta^{MAP})^2} + \frac{n_0}{(1-\theta^{MAP})^2} =$$
$$\frac{12}{(-9+\sqrt{93})^2} + \frac{28}{(11-\sqrt{93})^2} \Rightarrow A^{-1} = \frac{155-16\sqrt{93}}{31}.$$

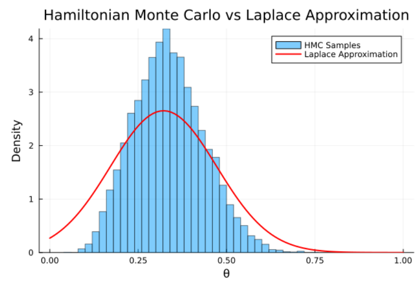
Thus, the Laplace approximation gives

$$\Theta|X \sim \mathcal{N}\left(\frac{\sqrt{93}-9}{2}, \frac{155-16\sqrt{93}}{31}\right).$$

By scaling the true posterior with $\frac{1}{\int p_X(x|\theta)\, p_\Theta(\theta)\, d\theta}$ and plotting the Laplace posterior approximation together with this, one can see it is a quite good approximation.



(a) Plot of the true posterior together with Laplace posterior approximation



(b) Plot of the Hamiltonian Monte Carlo approximation, with acceptance rate 81% using 10000 samples, together with Laplace posterior approximation